
Acceptance of self-driving cars

Description

In a questionnaire study we surveyed the citizens of two large universities about their attitudes regarding self-driving vehicles, their opinions about technological innovations in general and the willingness to use/buy self-driving cars. Our goal was to examine the relationship between general attitudes toward technological innovations, attitudes toward self-driving vehicles and acceptance of self-driving cars.

Attitudes toward self-driving vehicles was measured using the *Autonomous Vehicle Acceptability Scale (AVAS)*. It consists of four subscales:

- *benefits in usefulness* (eg. “A self-driving car would lower vehicle emissions.”)
- *benefits in situations* (eg. “A self-driving car could transport me even when I’m drunk, impaired or under medication.”)
- *commonalities concerns* (eg. “Public transportation such as buses that are completely self-driving.”)
- *system concerns* (eg. “A self-driving car may require additional learning by the driver to operate it.”)

The answers were given on seven-point Likert scales ranging from “Totally disagree” to “Totally agree” for the first two subscales, and ranging from “I don’t worry at all” to “I’m very worried” for the last two subscales. Higher scores on the first and second subscales imply a positive attitude, whereas higher scores on the third and fourth subscales a negative attitude towards self-driving vehicles.

Opinions and experiences about technological innovations in general were measured using the Technology Adoption Propensity (TAP) scale, which consists of three subscales:

- *proficiency* (eg. “I seem to have fewer problems than other people in making technology work.”)
- *dependency* (eg. “I feel like I am overly dependent on technology.”)
- *optimism* (eg. “New technologies make my life easier.”)

Answers could be given on a seven-point Likert scale ranging from “I strongly disagree” to “I strongly agree”. Higher scores on the subscales imply greater perceived proficiency, dependency and optimism accordingly.

We used three questions to measure acceptance of self-driving cars:

- “Would you test-drive a self-driving car?”
- “Would you drive a self-driving car in real life?”
- “Would you buy a self-driving car for your own use (regardless of your finances)?”

Answers were given on a seven-point Likert scale, ranging from “Definitely not” to “Definitely yes”. The average of the answers to these three questions were considered the measure of acceptance.

Additional questions regarded the age and gender of participants.

(Source: own collection – special subset of the original dataset)

Description of variables in the dataset

Variable	Short description
gender	Gender of participant (female/male)
age	Age of participant in years
biu	<i>benefits in usefulness</i> subscale of AVAS (1-7; mean score)
coc	<i>commonalities concerns</i> subscale of AVAS (1-7; mean score)
bis	<i>benefits in situations</i> subscale of AVAS (1-7; mean score)
syc	<i>system concerns</i> subscale of AVAS (1-7; mean score)
proficiency	<i>proficiency</i> subscale of TAP (1-7; mean score)
dependency	<i>dependency</i> subscale of TAP (1-7; mean score)
optimism	<i>optimism</i> subscale of TAP (1-7; mean score)
acceptance	measure of <i>acceptance</i> (1-7; mean score)

Dataset file: *selfdriving_acceptance.csv*